SAMIK BOSE

Fixed-Term Assistant Professor | Computational Molecular Science & Biophysics
Department of Computational Mathematics, Science and Engineering, Michigan State University
East Lansing, MI | bosesami@msu.edu | samikbose.github.io | ORCID: 0000-0002-0273-9162

ACADEMIC PROFILE

Computational scientist developing and implementing physics-based and machine learning-driven molecular simulation methods to resolve long-timescale molecular processes, reactive rare events, and kinetics. Research interests span across methods like weighted ensemble and Markov state modeling; QM/MM, ML/MM and ab initio molecular dynamics and AI-driven molecular generation to augment the understanding of molecular recognition, allosteric modulation and drug discovery. My research program is designed to connect rigorous statistical mechanical and electronic structure methods with structural biology, bio/chemistry, and medicinal chemistry, while mentoring students in an inclusive, equitable research environment. As an instructor, I continue to utilize evidence-based teaching pedagogies to enhance student learning in science and engineering courses.

ACADEMIC APPOINTMENTS

Fixed-Term Assistant Professor — **Aug 2024 – present**
Department of Computational Mathematics, Science and Engineering (CMSE), Michigan State University

Postdoctoral Research Associate — **Jan 2021 – Aug 2024**
Department of Biochemistry and Molecular Biology, Michigan State University | Mentor: Alex Dickson

EDUCATION

Ph.D., Theoretical and Computational Chemistry — **2014 – 2020**
Indian Association for the Cultivation of Science, India | Advisor: Debashree Ghosh
Dissertation: Polarization in Hybrid Quantum Mechanics/Molecular Mechanics Methods - Importance and Developments

M.Sc., Chemistry — **2012 – 2014**
University of Hyderabad, India | CGPA: 8.0
Dissertation topic: Thermal fluctuation and dynamics in a 2-D Frenkel-Kontorova chain. Relevant coursework: Statistical Mechanics & Thermodynamics, Electronic Structure, Spectroscopy, Chemical Kinetics.

B.Sc., Chemistry — **2009 – 2012**
St. Xavier's College, India | Grade: A
Relevant coursework: Thermodynamics, Bioinorganic Chemistry, Quantum Mechanics, Chemical Kinetics.

RESEARCH FUNDING

- **NIH/NIAID R21 (as co-investigator)** - $413,092. Molecular Mechanisms of CbrA Transceptor Signaling in Pseudomonas. PI: Benjamin J. Orlando, Michigan State University. (I am sole computational co-investigator)
- **MSU Research Foundation (MSURF) Postdoctoral Recruitment Grant** - $150,000. Award supporting recruitment and training of a postdoctoral researcher based on my submitted research proposal. (co-PI with Dr. Alex Dickson)
- **MSU Office of Undergraduate Research** - $12,000. Competitive support to train two undergraduate researchers for 10 months. (sole PI)

FUTURE RESEARCH DIRECTIONS

- Drug binding across (un)druggable protein targets with **reversible covalent inhibitors**: Explaining binding-to-reaction mechanism, kinetics, free energies and specificity by enhanced sampling QM/MM- and ML/MM molecular simulations.
- Deciphering the biophysical basis of kinetics- and dynamics-driven **allosteric modulation** for metallopeptidases and transmembrane proteins combining weighted-ensemble enhanced sampling with Markov-state modeling.
- Exploring **non-equilibrium fluctuations** in protein conformational space and their relevance in drug discovery using generative-AI and collective variable-driven MD.
- Develop molecularly learned reaction-field models that capture **self-consistent solute-solvent polarization** while retaining the efficiency and transferability of continuum embedding.

PEER-REVIEWED PUBLICATIONS

*Corresponding author; †equal contribution.

1. M. A. Orlando†, T. Shah†, M. W. Faber, **S. Bose***, B. J. Orlando*. Structure and conformational dynamics of the Pseudomonas CbrA transceptor. Protein Science 35(9), e70775 (2026). doi:10.1002/pro.70775

2. H. E. S. ElZorkany, H. Kandil, S. Jayaraman, A. Aly, S. H. Esfahani, D. Patel, D. Dannecker, M. Maciag, A. Paul, K. Lowran, S. Kumari, **S. Bose**, D. A. Ostrov, C. G. Wu, A. Dickson, T. J. Abbruscato, P. C. Trippier, B. J. Orlando, V. T. Karamyan. Discovery of a pyridine-piperazine-based small molecule that enhances the activity of peptidase neurolysin. Journal of Pharmacology and Experimental Therapeutics 393, 103827 (2026).

3. S. Bose, C. Kilinc, A. Dickson. Markov State Models with Weighted Ensemble Simulation: How to Eliminate the Trajectory Merging Bias. Journal of Chemical Theory and Computation 21, 1805–1816 (2025).

4. S. Bose, S. D. Lotz, I. Deb, M. Schuck, K. S. S. Lee, A. Dickson. How Robust is Ligand Binding Transition State? Journal of the American Chemical Society 145, 25318–25331 (2023).

5. N. Donyapour, F. F. Niazi, N. Roussey, **S. Bose**, A. Dickson. Flexible Topology: A New Method for Dynamic Drug Design. Journal of Chemical Theory and Computation 19, 5088–5098 (2023).

6. S. Bose, S. Chakrabarty, D. Ghosh. Support Vector Regression Based Monte Carlo Simulation of Flexible Water Clusters. ACS Omega 5, 7065–7073 (2020).

7. S. Bose, D. Dhawan, S. Nandi, R. R. Sarkar, D. Ghosh. Machine Learning Prediction of Interaction Energies in Rigid Water Clusters. Physical Chemistry Chemical Physics 20, 22987–22996 (2018).

8. R. Chakraborty†, **S. Bose**†, D. Ghosh. Effect of Solvation on the Ionization of Guanine Nucleotide: A Hybrid QM/EFP Study. Journal of Computational Chemistry 38, 2528–2537 (2017).

9. S. Bose, D. Ghosh. An Interaction Energy Driven Biased Sampling Technique: A Faster Route to Ionization Spectra in Condensed Phase. Journal of Computational Chemistry 38, 2248–2257 (2017).

10. S. Bose, S. Chakrabarty, D. Ghosh. Electrostatic Origin of the Red Solvatochromic Shift of DFHBDI in RNA Spinach. Journal of Physical Chemistry B 121, 4790–4798 (2017).

11. S. Bose, S. Chakrabarty, D. Ghosh. Effect of Solvation on Electron Detachment and Excitation Energies of a Green Fluorescent Protein Chromophore Variant. Journal of Physical Chemistry B 120, 4410–4420 (2016).

MANUSCRIPTS & PREPRINTS

1. K. M. Babin, C. Kilinc, **S. Bose**, S. E. Gostynska, A. Dickson, A. A. Pioszak. An allosteric CLR N-terminal swing-out mechanism for RAMP-mediated adrenomedullin receptor selectivity. Journal of the American Chemical Society (in revision)

2. S. Bose*†, N. Emoett†, N. French, A. Gauli, A. Dickson. Weighted-Ensemble Ab Initio Molecular Dynamics for Reactive Pathway Discovery and Kinetic Estimation. Manuscript in preparation. *Corresponding author.

3. S. Bose, A. Aly, V. T. Karamyan, B. J. Orlando, A. Dickson. Conformation Driven Enhancement of Neurolysin Activity in Presence of a Small Molecule Activator. bioRxiv (2026). doi:10.64898/2026.01.05.697776.

SELECTED RESEARCH ACCOMPLISHMENTS

- Developed and applied weighted ensemble-based enhanced sampling methods to explain kinetics driven drug discovery in long timescale biomolecular (un)binding with up to 10^{10} times speed-up.
- Developed an enhanced sampling method integrating Markov state models with weighted ensemble MD, two conceptually distinct sampling strategies, to improve robustness and accuracy of (un)binding kinetics.
- Applied machine learning to flexible-topology simulations for generation of drug-like small-molecule inhibitors within protein binding pockets guided by free-energy landscapes.
- Construction of polarizable water models for accurate interaction energies.
- Deciphered the binding site and mechanism of an allosteric small-molecule activator of the clinically relevant metallopeptidase neurolysin through computational modeling and simulation.
- Investigated the biophysics of small-molecule transport and signaling through transmembrane proteins using enhanced-sampling simulations and machine-learning-based predictive tools.
- Developed a biased-sampling algorithm for condensed-phase chromophore spectral features at ~10^3-fold lower computational cost.
- Decoded environmental effects on GFP-chromophore electronic structure and RNA-induced spectral shifts using multiscale molecular modeling.

TEACHING & MENTORING EXPERIENCE

Fixed-Term Assistant Professor, Michigan State University

- Taught **Computational Medicine** (400/800-level, Lead Instructor, Fall 2024 and Fall 2026): Designed a new curriculum for ~25 undergraduate and ~5 graduate students and implemented flipped-classroom pedagogy.
- Taught **Linear Algebra and Matrix Applications** (300-level, Instructor, Spring and Fall 2025): Three sections of ~60 undergraduates each; co-developed evaluation rubrics and course structure.
- Nominated by the college for a university-level **teaching excellence award**, April 2026.
- Selected for an **independent mentorship** of an REU student (May-July 2026) through research proposal.

Postdoctoral Research Associate, Michigan State University

- **Machine Learning for Molecular Dynamics** (900-level course, Guest Instructor, Spring 2023): Developed two lecture modules and two laboratory modules for ~20 graduate students.
- Mentored six undergraduate research assistants over 6-8 semesters, culminating in multiple poster presentations.
- Mentored graduate students in the Dickson Laboratory toward their Ph.D. research goals.

Doctoral Training, Indian Association for the Cultivation of Science

- Mentored approximately six undergraduate researchers/trainees, culminating in project presentations and posters.
- Mentored a C. V. Raman exchange fellow from Nigeria in quantum-chemistry methods, contributing to an international science and diversity exchange program.

TECHNICAL SKILLS & RESEARCH ASSETS

- Scientific software development and version control: Developer of wepy, CSNAnalysis, MLForce, MBC-MSM, Flexible Topology, and PartQMMM (all maintained in git).
- Programming: 10+ years of scientific programming in Python, R, Fortran, and C and associated computing libraries.
- Molecular simulation and enhanced sampling: weighted-ensemble simulation, Markov-state models, classical molecular dynamics, QM/MM dynamics, and Born-Oppenheimer / ab initio molecular dynamics.
- Molecular-modeling platforms: OpenMM, AMBER, Tinker, CHARMM, CHARMM-GUI, GROMACS, PySCF, Q-Chem, Molpro, and Gaussian.
- Machine learning: diffusion models, convolutional neural networks, support-vector machines, clustering, dimensionality reduction, and workflows in PyTorch, scikit-learn, and kernlab (R).
- Computer-aided drug discovery: small-molecule virtual screening and lead-optimization workflows using AutoDock, Schrödinger, and related tools.

- Data-intensive chemistry and drug discovery: experience with ChEMBL, GDB-13, QM9, and related molecular datasets.
- Collaborative research: extensive work with experimental and computational collaborators including cryo-EM specialists, structural biologists, spectroscopists, biochemists, and medicinal chemists; independently driving collaborative projects to resolve biomolecular mechanism questions.

PROFESSIONAL DEVELOPMENT & TEACHING INNOVATION

- CIRTL Network training: An Introduction to Evidence-Based Undergraduate STEM Teaching.
- Ongoing Certification in College Teaching through Michigan State University Graduate Educator Advancement and Teaching Office.
- Team lead, MSU-CMSE: community-driven generative-AI guidelines for students and teaching, including development of a specialized rubric for robust GenAI inclusion.

ORAL & POSTER PRESENTATIONS (SELECTED)

- Poster presentations — ACS Fall 2026, Chicago, IL: NVIDIA GPU Award finalist; BIOL Division Sci-Mix.
- Oral presentation — ACS Fall 2025, COMP Division, Washington, DC.
- Poster presentation — Midwest Theoretical and Computational Chemistry Symposium, May 2025, Wayne State University.
- Poster presentation — J-1 Scholar Showcase, April 2025, Michigan State University.
- Poster presentation — J-1 Scholar Showcase, April 2024, Michigan State University.
- Poster presentation — Biochemistry and Molecular Biology Scientific Symposium, May 2024, Michigan State University.
- Oral presentation — ACS Spring 2024, COMP Division, New Orleans, LA.
- Oral presentation — ACS Spring 2024, BIOT Division, New Orleans, LA.
- Oral presentation — Postdoctoral Symposium, September 2023, Michigan State University.
- Poster presentation — Midwest Undergraduate Computational Chemistry Consortium, August 2023, Michigan State University.
- Oral presentation — MD Simulation: Focus on Methods, December 2022, TIFR Hyderabad, India.
- Oral presentation — ACS Fall 2022, COMP Division, Chicago, IL.
- Oral presentation — Midwest Protein Folding, May 2022, University of Notre Dame.
- Oral presentation — ACS Spring 2022, COMP Division (virtual).
- Oral presentation — Molecular Biophysics Supergroup, February 2022, Michigan State University.
- Oral presentation — Research Scholars Outreach, January 2020, NCST, Kolkata, India.
- Oral presentation — Physical Chemistry Symposium, October 2019, Indian Association for the Cultivation of Science, Kolkata, India.

AWARDS & FELLOWSHIPS

- Nominated by the college for a university-level teaching excellence award, Michigan State University, April 2026.
- CSIR-UGC National Eligibility Test (NET), All-India Rank 46 in Chemistry, May 2014; full Ph.D. fellowship support.
- Graduate Aptitude Test in Engineering (GATE), All-India Rank 214 in Chemistry, May 2014.
- University of Hyderabad M.Sc. Entrance Examination, All-India Rank 5, 2012.
- University of Delhi M.Sc. Entrance Examination, All-India Rank 3, 2012.
- DST-INSPIRE National Scholarship, 2009–2014; support through B.Sc. and M.Sc. training.

PROFESSIONAL MEMBERSHIPS

- **American Chemical Society (ACS),** including the COMP Division, BIOL Division and local section.
- **Biophysical Society (BPS)** and its Theory & Computation subgroup.
- MichBio community member; participation in events focused on biomedical advances.

SERVICE & OUTREACH

- American Chemical Society service: Presided over oral-presentation sessions at ACS meetings (2022–2026) and contributed to community building among grad students and postdocs.
- Peer reviewer for PNAS, Journal of Computational Chemistry, Royal Society Open Science, LiveCoMS, and other scientific journals.
- Member of CMSE Social Events Committee, Michigan State University; Contributed to departmental community and bonding activities.
- Member of Instructor Search Committees, Department of Biochemistry and Molecular Biology, Michigan State University.
- Member of Team-UP Postdoctoral Search Committee, Department of Biochemistry and Molecular Biology, Michigan State University.
- Evaluator for UURAF, Mid-SURE, and related Michigan undergraduate-research events (2022-2026), supporting undergraduate research and outreach.
- Scientific outreach and collaboration-building activities in India, including facilitation of inter-laboratory collaborations.
- Team lead, CMSE generative-AI teaching initiative: community-driven guidelines and rubric development for responsible classroom use.

REFERENCES

Alex Dickson, Ph.D. - Professor
Departments of Biochemistry and Molecular Biology; Computational Mathematics, Science and Engineering
Michigan State University, East Lansing, MI 48824
Email: alexrd@egr.msu.edu

Debashree Ghosh, Ph.D. - Professor
School of Chemical Sciences
Indian Association for the Cultivation of Science, Kolkata 700032, West Bengal, India
Email: pcdg@iacs.res.in | debashreeghosh@gmail.com

Benjamin J. Orlando, Ph.D. - Associate Professor
Department of Biochemistry and Molecular Biology
Michigan State University, East Lansing, MI 48824
Email: orlandob@msu.edu

Augen A. Pioszak, Ph.D. - Associate Professor
Department of Biochemistry and Physiology
University of Oklahoma Health Sciences Center, Oklahoma City, OK 73126
Email: augen-pioszak@ou.edu

Vardan T. Karamyan, Ph.D. - Professor
Department of Foundational Medical Studies
Oakland University William Beaumont School of Medicine, Rochester, MI 48309
Email: vkaramyan@oakland.edu

Devin Silvia, Ph.D. - Director of Undergraduate Studies (Teaching Reference)
Department of Computational Mathematics, Science and Engineering
Michigan State University, East Lansing, MI 48824
Email: dsilvia@msu.edu